TU WIEN Informatics

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Techniques for Dynamic Visualization of Time-Varying Flood Data

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Problem Statement

Flood events, change over time and are difficult for the general public to interpret. Existing flood maps often do not clearly communicate extent and likelihood. There is a need for visualizations that are easy to understand, accessible, and informative for non-expert users, helping them make safe decisions in affected areas.

For this project Global Flood Monitoring (GFM) data from Copernicus of a flood in Thessaly Greece (2018) was used.

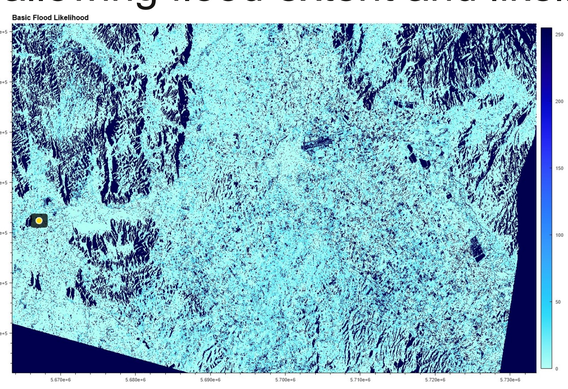
Aim of the Work

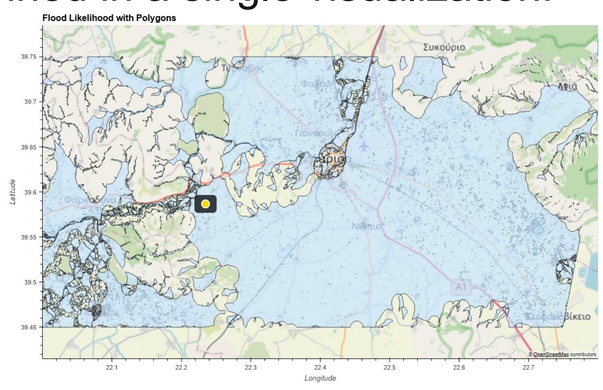
Develop visualizations of time-varying flood events using satellite data

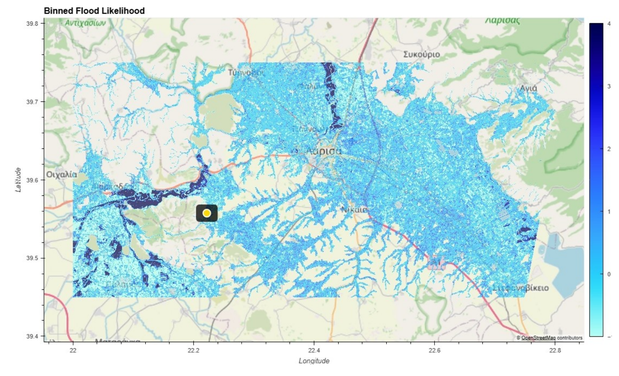
- Explore and implement new visualization techniques
- Communicate flood extent and likelihood effectively
- Conduct a survey on different visualization techniques
- Evaluate visualization on survy result

Development and Implementation of Visualizations

During the development of the visualizations, several different approaches were tested to improve interpretability. A baseline visualization that simply plotted the raw data was created first. One approach involved using polygons to smooth the plotted areas, making the flood shapes easier to interpret. A background map was also added to help users better understand the geographic context. Data binning was implemented to reduce complexity and highlight meaningful patterns more clearly, allowing flood extent and likelihood to be combined in a single visualization.





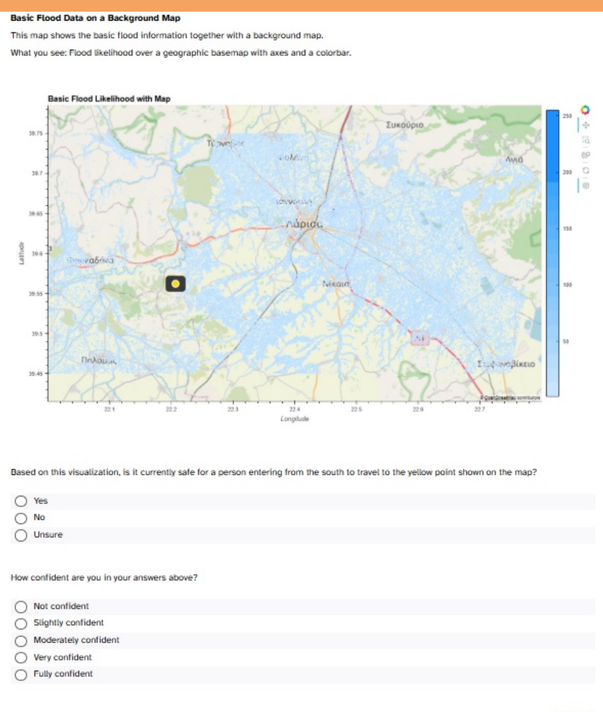


Survey

A user survey was conducted to evaluate how well the developed visualizations communicate flood information to the general public. Participants were asked to complete simple interpretation tasks. Their feedback helped identify which approaches were easiest to understand and most intuitive for non-experts.

- Participants evaluated multiple visualization versions (baseline + new designs)
- Task-based questions measured accuracy and confidence.
- Users compared visualizations and rated their preferences.
- Survey included accessibility aspects (color-blind-friendly design)

The survey was done with the help of SoSci Survey (www.soscisurvey.de)



Result